Signals and sampling

- A continuous signal x(t) is sampled every T seconds.

- The sampling rate is f_s = 1 / T.

- Nyquist: f_s must exceed twice the highest frequency.

- Below it, frequencies fold back: aliasing.



Quantization

- Each sample is rounded to one of 2^N levels.
- The step is the full range divided by 2^N.
- Quantization noise is about 6 dB lower per bit.
- Dither spreads the error into broadband noise.

Helvetica: the default box.

Georgia, bold, on a filled box with a frame.

Courier New, italic and underlined

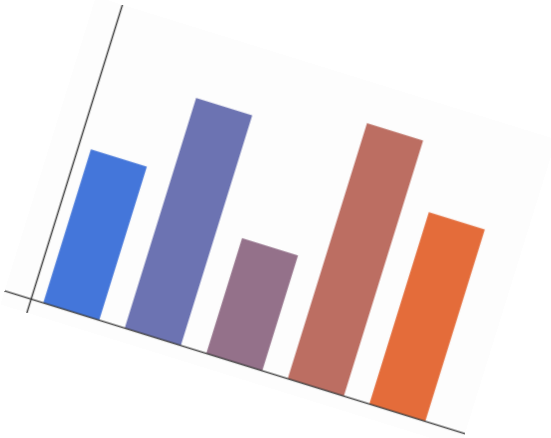
Avenir Next, turned

Read more on Wikipedia

The z-transform



- X(z) is the sum of x[n] z^-n over every n.
- A delay of one sample multiplies by z^-1.
- Poles inside the unit circle mean a stable filter.
- ~~Zeros on the circle~~ remove a frequency.



Filters in practice

- FIR filters are always stable and can have linear phase.

- IIR filters reach a steep edge with fewer taps.

- Windowing trades the main lobe for side lobes.

- Exercise: design a 5-tap moving average.

